Problem Set 2: Kinematics

Problem 1

A cart starts from rest and speeds up at a steady rate. Its average velocity over a time interval is

$$\bar{v} = \frac{\Delta x}{\Delta t} = \frac{x_f - x_0}{t_f - t_0}. \tag{2.1}$$

(a) Find the average velocity between $t = 0$ and $t = 4.0$ s.

(b) Explain why the instantaneous velocity at $t = 2.0$ s equals it.

(c) Sketch the position of the cart against time.

Problem 2

Starting from the definition of acceleration, the speed after a distance d follows from three steps:

$$\begin{aligned} v^2 &= v_0^2 + 2ad \\ &= 0 + 2\left(3.0\ \text{m/s}^2\right)(12\ \text{m}) \\ &= 72\ \text{m}^2/\text{s}^2. \end{aligned}$$

Which value is closest to the final speed?

A. 6.0 m/s

B. 8.5 m/s

C. 36 m/s

D. 72 m/s

Problem 3

A stone falls from rest. Ignoring air resistance, the distance it falls in time t is

$$d = \frac{1}{2} g t^2, \tag{2.2}$$

and the time to fall a height h is

$$t = \sqrt{\frac{2h}{g}}. \tag{2.3}$$

Use $g = 9.8\ \text{m/s}^2$ in both parts:

(i) how far does it fall in 1.5 s?

(ii) how long does it take to fall 20 m?

2.5 metres is the height of the classroom ceiling, for comparison.